Derivation of the 1D Linear Diffusivity Equation

Convention: flow in the **−x direction**, mass balanced over a time interval **Δt**. Every step is written out.

Part 0: Setup and assumptions

Geometry: a rectangular block of porous rock with width w, height h, and flow only along x. The cross-sectional area is

$$A = w\,h$$

We study a thin slice between x and x + Δx. Its bulk volume is

$$\Delta V = w\,h\,\Delta x$$

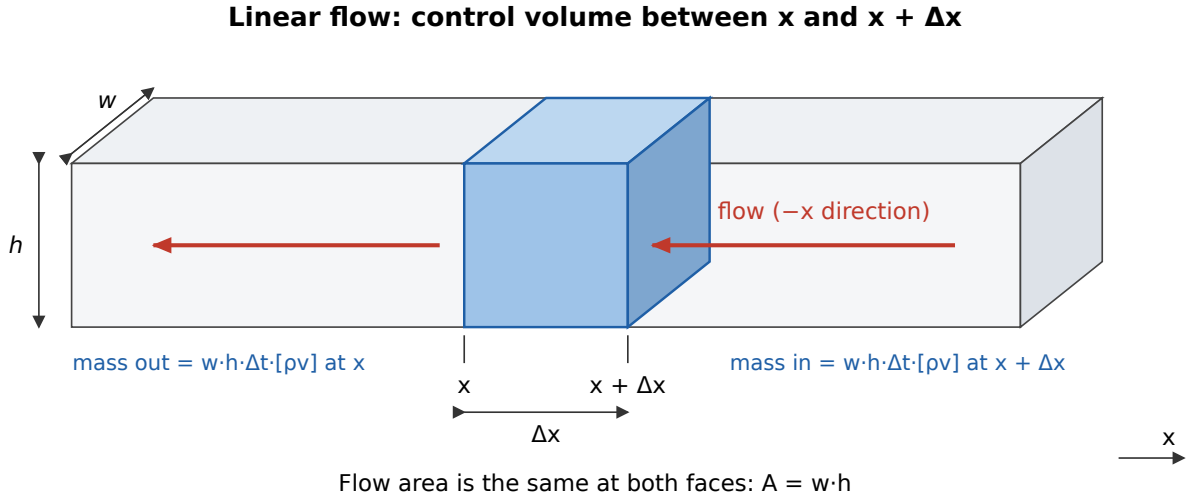


Figure 1. Linear flow. Fluid enters the slice at x + Δx and leaves at x. The flow area w·h is the same on both faces.

Assumptions:

1. Flow in one direction only (x)
2. Single-phase, slightly compressible fluid
3. Homogeneous rock: k constant. Fluid viscosity μ constant
4. Darcy's law applies
5. Horizontal flow (no gravity)
6. Isothermal
7. Small pressure gradients

Variables:

- ρ = fluid density
- v = Darcy velocity (volume flow rate per unit area)
- φ = porosity
- p = pressure
- c = fluid compressibility
- c_f = rock (pore) compressibility

Part 1: Mass of fluid crossing a face

Mass = density × volume. The volume of fluid crossing area A at velocity v during time Δt is A·v·Δt. Multiply by density:

$$\text{mass} = \rho\,A\,v\,\Delta t = \rho\,v\,w\,h\,\Delta t$$

Units check: (kg/m³)·(m/s)·m²·s = kg ✓

Part 2: Mass in, mass out, mass accumulated

Fluid moves in the −x direction. It **enters** the slice through the right face at x + Δx and **leaves** through the left face at x.

Mass in (through the face at x + Δx):

$$(\text{ mass })_{in} = w\,h\,\Delta t\,[\,\rho v\,]_{x+\Delta x}$$

Mass out (through the face at x):

$$(\text{ mass })_{out} = w\,h\,\Delta t\,[\,\rho v\,]_{x}$$

Mass accumulated inside the slice. The mass stored in the slice at any time is:

$$\text{mass stored} = (\text{ pore volume }) \times \rho = (\,\phi\,\Delta V)\;\rho = \Delta V\,(\,\phi\rho\,)$$

The accumulation is the change in stored mass between time t and t + Δt:

$$(\text{ mass })_{acc} = \Delta V[\,(\,\phi\rho\,)_{t+\Delta t} - (\,\phi\rho\,)_{t}]$$

Part 3: Mass balance

Conservation of mass says matter is neither created nor destroyed:

$$(\text{ mass })_{in} - (\text{ mass })_{out} = (\text{ mass })_{acc}$$

Substitute the three expressions:

$$w\,h\,\Delta t\,[\,\rho v\,]_{x+\Delta x} - w\,h\,\Delta t\,[\,\rho v\,]_{x} = w\,h\,\Delta x\,[\,(\,\phi\rho\,)_{t+\Delta t} - (\,\phi\rho\,)_{t}]$$

Factor w·h·Δt on the left:

$$w\,h\,\Delta t\,(\,[\,\rho v\,]_{x+\Delta x} - [\,\rho v\,]_{x}) = w\,h\,\Delta x\,[\,(\,\phi\rho\,)_{t+\Delta t} - (\,\phi\rho\,)_{t}]$$

Part 4: Divide and take limits

Divide both sides by w·h·Δx·Δt.

- Left side: w·h cancels, Δt cancels, leaving Δx in the denominator.
- Right side: w·h cancels, Δx cancels, leaving Δt in the denominator.

$$\frac{[\,\rho v\,]_{x+\Delta x} - [\,\rho v\,]_{x}}{\Delta x} = \frac{(\,\phi\rho\,)_{t+\Delta t} - (\,\phi\rho\,)_{t}}{\Delta t}$$

This is where Δt and w·h disappear from the equation.

Now let Δx → 0 and Δt → 0. By the definition of the derivative:

$$\lim_{\Delta x \to 0} \frac{f\,(\,x+\Delta x\,) - f\,(\,x\,)}{\Delta x} = \frac{\partial f}{\partial x}$$

So the left side becomes ∂(ρv)/∂x and the right side becomes ∂(φρ)/∂t. This is the continuity equation, eq. (1):

$$\frac{\partial\,(\,\rho v\,)}{\partial x} = \frac{\partial\,(\,\phi\rho\,)}{\partial t} \qquad (\,1\,)$$

Part 5: Darcy's law

Darcy's law says flow goes from high pressure to low pressure at a rate proportional to the pressure gradient:

$$v = -\frac{k}{\mu}\frac{\partial p}{\partial s}$$

where s is the direction of flow. Here the flow is in the **−x direction**, so s = −x and ∂/∂s = −∂/∂x:

$$v = -\frac{k}{\mu}\left(-\frac{\partial p}{\partial x}\right) = +\frac{k}{\mu}\frac{\partial p}{\partial x} \qquad (\,2\,)$$

Sanity check: if pressure increases with x, then ∂p/∂x > 0 and v > 0. Fluid moves from high pressure (large x) toward low pressure (small x), which is the −x direction ✓

Part 6: Combine continuity and Darcy

Substitute (2) into the left side of (1):

$$\rho v = \rho \frac{k}{\mu} \frac{\partial p}{\partial x}$$

$$\frac{\partial}{\partial x}\left(\rho \frac{k}{\mu} \frac{\partial p}{\partial x} \right) = \frac{\partial \left(\phi \rho \right)}{\partial t}$$

Since k and μ are constants, they can be pulled outside the derivative:

$$\frac{k}{\mu} \frac{\partial}{\partial x}\left(\rho \frac{\partial p}{\partial x} \right) = \frac{\partial \left(\phi \rho \right)}{\partial t} \qquad (3)$$

Now we must deal with ρ and φ, which both depend on pressure. For that we need compressibility.

Part 7: Compressibility definitions

Fluid compressibility: the fractional change in density per unit pressure change:

$$c = \frac{1}{\rho} \frac{\partial \rho}{\partial p}$$

Multiply both sides by ρ:

$$\frac{\partial \rho}{\partial p} = c \, \rho \qquad (4)$$

Rock compressibility: the fractional change in porosity per unit pressure change:

$$c_f = \frac{1}{\phi} \frac{\partial \phi}{\partial p}$$

$$\frac{\partial \phi}{\partial p} = c_f \, \phi \qquad (5)$$

Chain rule. Since ρ and φ depend on x and t only through p(x,t):

$$\frac{\partial \rho}{\partial x} = \frac{\partial \rho}{\partial p} \frac{\partial p}{\partial x} = c \, \rho \frac{\partial p}{\partial x} \qquad (6)$$

$$\frac{\partial \rho}{\partial t} = \frac{\partial \rho}{\partial p} \frac{\partial p}{\partial t} = c \, \rho \frac{\partial p}{\partial t} \qquad (7)$$

$$\frac{\partial \phi}{\partial t} = \frac{\partial \phi}{\partial p} \frac{\partial p}{\partial t} = c_f \, \phi \frac{\partial p}{\partial t} \qquad (8)$$

Part 8: Right-hand side of (3) (accumulation term)

Apply the product rule, ∂(ab)/∂t = a ∂b/∂t + b ∂a/∂t:

$$\frac{\partial \left(\phi \rho \right)}{\partial t} = \phi \frac{\partial \rho}{\partial t} + \rho \frac{\partial \phi}{\partial t}$$

Substitute (7) and (8):

$$= \phi \left(c \, \rho \frac{\partial p}{\partial t} \right) + \rho \left(c_f \, \phi \frac{\partial p}{\partial t} \right)$$

Factor out φ ρ ∂p/∂t:

$$= \phi \, \rho \, \left(c + c_f \right) \frac{\partial p}{\partial t}$$

Define **total compressibility** c_t = c + c_f:

$$\frac{\partial \left(\phi \rho \right)}{\partial t} = \phi \, \rho \, c_t \frac{\partial p}{\partial t} \qquad (9)$$

Part 9: Left-hand side of (3) (flow term)

Apply the product rule to ρ ∂p/∂x, treating ρ as a and ∂p/∂x as b:

$$\frac{\partial}{\partial x}\left(\rho \frac{\partial p}{\partial x} \right) = \rho \frac{\partial^2 p}{\partial x^2} + \frac{\partial p}{\partial x} \frac{\partial \rho}{\partial x}$$

Substitute (6), ∂ρ/∂x = c ρ ∂p/∂x:

$$= \rho \frac{\partial^2 p}{\partial x^2} + \frac{\partial p}{\partial x} \cdot c \rho \frac{\partial p}{\partial x} = \rho \frac{\partial^2 p}{\partial x^2} + c \rho \left(\frac{\partial p}{\partial x} \right)^2$$

Factor out ρ:

$$= \rho \left[\frac{\partial^2 p}{\partial x^2} + c \left(\frac{\partial p}{\partial x} \right)^2 \right] \qquad (10)$$

Part 10: Put both sides together

Substitute (10) and (9) into (3):

$$\frac{k}{\mu} \rho \left[\frac{\partial^2 p}{\partial x^2} + c \left(\frac{\partial p}{\partial x} \right)^2 \right] = \phi \rho c_t \frac{\partial p}{\partial t}$$

Divide both sides by ρ (it cancels):

$$\frac{k}{\mu} \left[\frac{\partial^2 p}{\partial x^2} + c \left(\frac{\partial p}{\partial x} \right)^2 \right] = \phi c_t \frac{\partial p}{\partial t}$$

Multiply both sides by μ/k:

$$\frac{\partial^2 p}{\partial x^2} + c \left(\frac{\partial p}{\partial x} \right)^2 = \frac{\phi \mu c_t}{k} \frac{\partial p}{\partial t} \qquad (11)$$

Equation (11) is **nonlinear** because of the squared gradient term, so it is hard to solve analytically.

Part 11: Linearization

We compare the two terms on the left of (11):

- Term A: ∂²p/∂x²
- Term B: c (∂p/∂x)²

For a slightly compressible liquid (oil, water), c is very small, typically 10^{-6} to 10^{-5} psi⁻¹. The pressure gradient is also small away from the wellbore. Term B contains c times a squared gradient, so it is far smaller than Term A.

Example: with c = 10^{-5} psi⁻¹ and ∂p/∂x = 10 psi/ft, Term B is $10^{-5} \times 100 = 10^{-3}$ psi/ft². Term A is usually much larger.

So we drop Term B, which gives the diffusivity equation:

$$\frac{\partial^2 p}{\partial x^2} = \frac{\phi \mu c_t}{k} \frac{\partial p}{\partial t} \qquad (12)$$

Part 12: Hydraulic diffusivity form

Define

$$\eta = \frac{k}{\phi \mu c_t}$$

Multiply both sides of (12) by η:

$$\frac{\partial p}{\partial t} = \eta \frac{\partial^2 p}{\partial x^2}$$

SI units check:

$$\eta = \frac{\mathrm{m}^2}{(\mathrm{Pa \cdot s}) \left(\mathrm{Pa}^{-1} \right)} = \frac{\mathrm{m}^2}{\mathrm{s}}$$

This matches the units of a diffusion coefficient ✓

Oilfield units: with k in md, μ in cp, c_t in psi⁻¹, t in hours:

$$\eta = 0.0002637 \frac{k}{\phi \mu c_t} \quad \left[\frac{\text{ft}^2}{\text{hr}}\right]$$

Summary of the logic

Step	What we did	Result
1	Mass in − mass out = mass accumulated	Balance over Δt
2	Divide by w·h·Δx·Δt, take limits	Continuity equation (1)
3	Darcy's law for flow in −x	v = (k/μ) ∂p/∂x
4	Substitute Darcy into continuity	Equation (3)
5	Use c and c_f to rewrite ∂(φρ)/∂t	φ ρ c_t ∂p/∂t
6	Expand ∂(ρ ∂p/∂x)/∂x	Gives the nonlinear c (∂p/∂x)² term
7	Neglect that term (small c, small gradient)	Linear diffusivity equation

Radial Flow Diffusivity Equation

Same convention as before: flow toward the well, i.e. in the **−r direction**, with mass balanced over a time interval **Δt**.

Note on notation: c_f is the **formation (rock)** compressibility, where f stands for formation, not fluid. Fluid compressibility carries a phase subscript in general (c_o, c_w, c_g). With a single fluid we write it simply as c.

Radial Step 0: Geometry and assumptions

Consider a cylindrical reservoir of thickness h with a well at the center. Take a thin **annular shell** between radius r and r + Δr. The assumptions are the same as in the linear case, except flow is purely radial.

The key difference from the linear case is that the flow area depends on r. The area of a cylindrical surface at radius r is

$$A(r) = 2\pi r h$$

So the area at r + Δr is larger than at r. In the linear case A was constant and cancelled.

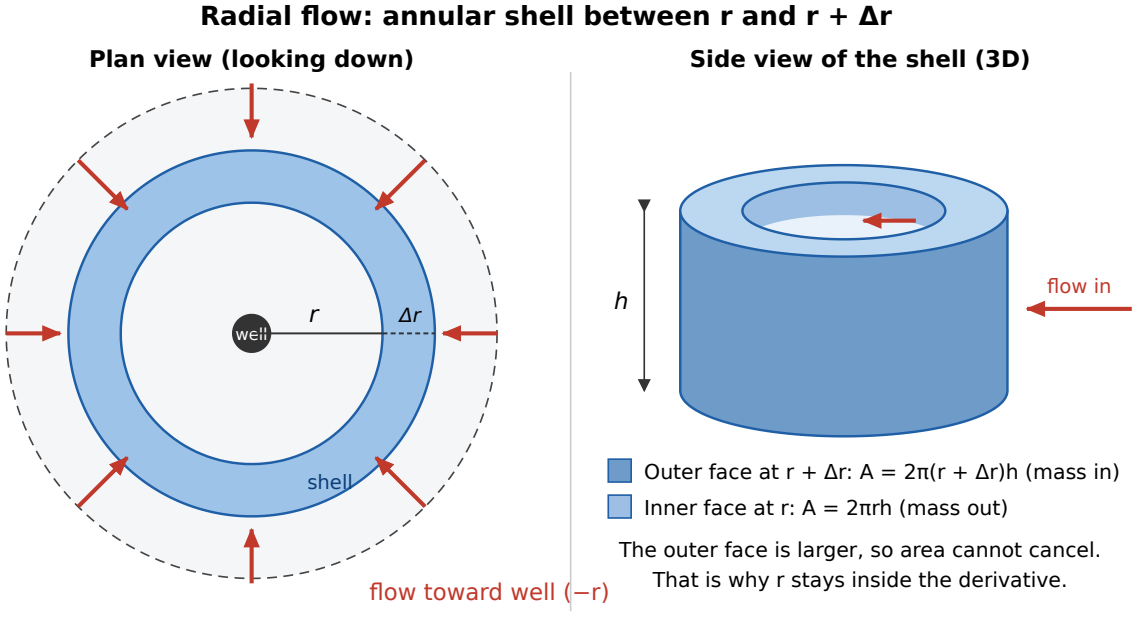


Figure 2. Radial flow. Left: plan view of the annular shell with flow converging on the well. Right: 3D view of the same shell. Fluid enters through the outer cylindrical face and leaves through the inner one, which has less area.

Radial Step 1: Mass crossing each face over Δt

Using mass = ρ A v Δt:

Mass in (outer face, at r + Δr):

$$(\text{ mass })_{in} = 2\pi h\,\Delta t\,[r\rho v]_{r+\Delta r}$$

Mass out (inner face, at r):

$$(\text{ mass })_{out} = 2\pi h\,\Delta t\,[r\rho v]_{r}$$

The radius r stays inside the bracket because it is part of the area, and the area changes between the two faces.

Radial Step 2: Mass accumulated in the shell

The bulk volume of the shell is the difference of two cylinders:

$$\Delta V = \pi h[\,(r+\Delta r)^2 - r^2] = \pi h(2r\,\Delta r + \Delta r^2)$$

When Δr is very small, Δr² is negligible compared with 2rΔr, so

$$\Delta V \approx 2\pi r h\,\Delta r$$

The accumulated mass is

$$(\text{ mass })_{acc} = 2\pi r h\,\Delta r[\,(\phi\rho)_{t+\Delta t} - (\phi\rho)_{t}]$$

Radial Step 3: Mass balance

(mass in) − (mass out) = (mass accumulated):

$$2\pi h\,\Delta t([\,r\rho v\,]_{r+\Delta r} - [\,r\rho v\,]_{r}) = 2\pi r h\,\Delta r[\,(\phi\rho)_{t+\Delta t} - (\phi\rho)_{t}]$$

Radial Step 4: Divide and take limits

Divide both sides by 2π h Δr Δt:

$$\frac{[\,r\rho v\,]_{r+\Delta r} - [\,r\rho v\,]_{r}}{\Delta r} = r\,\frac{(\phi\rho)_{t+\Delta t} - (\phi\rho)_{t}}{\Delta t}$$

Take Δr → 0 and Δt → 0:

$$\frac{\partial(r\rho v)}{\partial r} = r\,\frac{\partial(\phi\rho)}{\partial t}$$

Divide by r to get the **radial continuity equation**:

$$\frac{1}{r}\frac{\partial(r\rho v)}{\partial r} = \frac{\partial(\phi\rho)}{\partial t} \qquad (R1)$$

Compare with the linear case, ∂(ρv)/∂x = ∂(φρ)/∂t. The only change is the factor r inside the derivative, which comes from the growing area.

Radial Step 5: Darcy's law

For flow in the −r direction, as in the linear case:

$$v = +\frac{k}{\mu}\frac{\partial p}{\partial r} \qquad (R2)$$

Pressure increases with r, so fluid moves inward from high to low pressure ✓

Radial Step 6: Combine continuity and Darcy

Substitute (R2) into (R1) and pull the constants k and μ outside:

$$\frac{k}{\mu}\frac{1}{r}\frac{\partial}{\partial r}\left(r\rho\frac{\partial p}{\partial r}\right) = \frac{\partial(\phi\rho)}{\partial t} \qquad (R3)$$

Radial Step 7: Right-hand side (accumulation)

This is identical to the linear case, because it only involves time derivatives and compressibility:

$$\frac{\partial(\phi\rho)}{\partial t} = \phi\rho\,c_t\,\frac{\partial p}{\partial t} \qquad (R4)$$

where c_t = c + c_f.

Radial Step 8: Left-hand side (flow term)

We need the derivative of a product of **three** factors: r, ρ and ∂p/∂r. The product rule gives:

$$\frac{\partial}{\partial r}\left(r\rho\frac{\partial p}{\partial r}\right) = \rho\frac{\partial p}{\partial r} + r\frac{\partial p}{\partial r}\frac{\partial \rho}{\partial r} + r\rho\frac{\partial^2 p}{\partial r^2}$$

Use the chain rule with the compressibility definition, ∂ρ/∂r = c ρ ∂p/∂r:

$$= \rho\frac{\partial p}{\partial r} + r\,c\rho\left(\frac{\partial p}{\partial r}\right)^2 + r\rho\frac{\partial^2 p}{\partial r^2}$$

Now multiply by 1/r and factor out ρ:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\rho\frac{\partial p}{\partial r}\right) = \rho\left[\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r} + c\left(\frac{\partial p}{\partial r}\right)^2\right] \qquad (R5)$$

The term (1/r) ∂p/∂r is new. It exists only because of the radial geometry.

Radial Step 9: Combine both sides

Substitute (R5) and (R4) into (R3), cancel ρ, and multiply by μ/k:

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r} + c\left(\frac{\partial p}{\partial r}\right)^2 = \frac{\phi\mu c_t}{k}\frac{\partial p}{\partial t} \qquad (R6)$$

This is nonlinear because of the squared gradient term.

Radial Step 10: Linearization

Drop c (∂p/∂r)², since c is small (10^{-6} to 10^{-5} psi⁻¹) and the gradient is moderate:

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r} = \frac{\phi\mu c_t}{k}\frac{\partial p}{\partial t}$$

The left side is exactly (1/r) ∂/∂r (r ∂p/∂r). Expanding that product gives back the two terms. So the compact form is:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right) = \frac{1}{\eta}\frac{\partial p}{\partial t}, \qquad \eta = \frac{k}{\phi\mu c_t} \qquad (R7)$$

Caution: the gradient is largest near the wellbore, so the neglected term is least accurate there at high drawdown. It is still the standard assumption behind most well-test analysis.

Oilfield units: with r in ft, t in hours, k in md, μ in cp and c_t in psi⁻¹:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right) = \frac{\phi\mu c_t}{0.0002637\,k}\frac{\partial p}{\partial t}$$

Linear vs radial

	Linear	Radial
Flow area	constant (w·h)	2πrh, grows with r
Shell volume	w·h·Δx	about 2πrh·Δr
Continuity	∂(ρv)/∂x = ∂(φρ)/∂t	(1/r) ∂(rρv)/∂r = ∂(φρ)/∂t
Diffusivity equation	∂²p/∂x² = (1/η) ∂p/∂t	∂²p/∂r² + (1/r) ∂p/∂r = (1/η) ∂p/∂t

If the flow were outward (injection), the sign of v in Darcy's law and the sign in the continuity equation both flip, and they cancel. The final equation is the same.

Line-Source Solution (Exponential Integral)

We now solve the radial diffusivity equation (R7) for the most important practical case: a well producing at constant rate in an infinitely large reservoir.

Solution Step 1: Problem statement

Additional assumptions: the well is a line source (zero radius), constant rate q, infinite-acting reservoir, uniform initial pressure p_i, single-phase flow, constant properties.

PDE:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right)=\frac{\phi\mu c_t}{k}\frac{\partial p}{\partial t}$$

Initial condition (uniform pressure):

$$p\left(r,0\right)=p_i$$

Outer boundary (infinite reservoir, pressure far away is undisturbed):

$$p\left(r\to\infty,t\right)=p_i$$

Inner boundary (constant rate q at the line source). Darcy's law at the well gives q = (2π k h/μ) r ∂p/∂r in SI units, so:

$$\lim_{r\to 0}\ r\frac{\partial p}{\partial r}=\frac{q\,\mu}{2\pi kh}$$

Solution Step 2: Dimensionless variables

In oilfield units (q in STB/D, B in RB/STB, r_w = wellbore radius):

$$p_D=\frac{kh\,\left(p_i-p\right)}{141.2\,qB\mu},\qquad r_D=\frac{r}{r_w},\qquad t_D=\frac{0.0002637\,k\,t}{\phi\mu c_t r_w^2}$$

The PDE becomes parameter-free:

$$\frac{1}{r_D}\frac{\partial}{\partial r_D}\left(r_D\frac{\partial p_D}{\partial r_D}\right)=\frac{\partial p_D}{\partial t_D}$$

The conditions become:

$$p_D\left(r_D,0\right)=0,\qquad p_D\left(r_D\to\infty,t_D\right)=0,\qquad \lim_{r_D\to 0}\ r_D\frac{\partial p_D}{\partial r_D}=-1$$

The minus sign appears because p_D is defined with (p_i − p), so it decreases with r.

Solution Step 3: Boltzmann transformation

The two variables r_D and t_D can be combined into one. Define

$$s=\frac{r_D^2}{4\,t_D}$$

and look for a solution of the form p_D = f(s). We need the derivatives of s:

$$\frac{\partial s}{\partial r_D}=\frac{r_D}{2t_D}=\frac{2s}{r_D},\qquad \frac{\partial s}{\partial t_D}=-\frac{r_D^2}{4t_D^2}=-\frac{s}{t_D}$$

Right side (time derivative), by the chain rule:

$$\frac{\partial p_D}{\partial t_D}=f'\left(s\right)\frac{\partial s}{\partial t_D}=-\frac{s}{t_D}f'$$

Left side (space derivatives). First the inner derivative:

$$\frac{\partial p_D}{\partial r_D}=f'\frac{r_D}{2t_D}$$

Multiply by r_D:

$$r_D\frac{\partial p_D}{\partial r_D}=f'\frac{r_D^2}{2t_D}=2sf'$$

Differentiate with respect to r_D, using the chain rule again (d/ds of 2sf' is 2f' + 2sf''):

$$\frac{\partial}{\partial r_D}\left(r_D\frac{\partial p_D}{\partial r_D}\right)=(2f'+2sf'')\frac{r_D}{2t_D}=(f'+sf'')\frac{r_D}{t_D}$$

Multiply by 1/r_D:

$$\frac{1}{r_D}\frac{\partial}{\partial r_D}\left(r_D\frac{\partial p_D}{\partial r_D}\right)=\frac{f'+sf''}{t_D}$$

Set left = right. The factor 1/t_D cancels, so the PDE has become an ordinary differential equation in s only:

$$f' + sf'' = -sf' \quad \Rightarrow \quad sf'' + (1+s)\,f' = 0$$

Solution Step 4: Solve the ODE

Let g = f'. Then g' = f'' and

$$sg' = -(1+s)\,g \quad \Rightarrow \quad \frac{g'}{g} = -\frac{1}{s} - 1$$

Integrate both sides:

$$\ln g = -\ln s - s + C$$

$$g = f'(s) = C_1 \frac{e^{-s}}{s}$$

Solution Step 5: Apply the inner boundary condition

From Step 3, r_D ∂p_D/∂r_D = 2 s f'. Substitute f':

$$r_D \frac{\partial p_D}{\partial r_D} = 2s\,C_1 \frac{e^{-s}}{s} = 2C_1 e^{-s}$$

As r_D → 0, s → 0 and e^(−s) → 1, so the condition r_D ∂p_D/∂r_D → −1 gives 2C₁ = −1:

$$C_1 = -\frac{1}{2} \quad \Rightarrow \quad f'(s) = -\frac{e^{-s}}{2s}$$

Solution Step 6: Apply the outer boundary and initial conditions

Both the initial condition (t_D = 0) and the outer boundary (r_D → ∞) correspond to s → ∞, where p_D = 0. So f(∞) = 0 and the same condition handles both. Integrate f' from s to ∞:

$$f(\infty) - f(s) = \int_s^\infty f'(u)\,du = -\frac{1}{2}\int_s^\infty \frac{e^{-u}}{u}\,du$$

Since f(∞) = 0:

$$f(s) = \frac{1}{2}\int_s^\infty \frac{e^{-u}}{u}\,du$$

This integral is the **exponential integral** E₁(s), also written −Ei(−s). So:

$$p_D(r_D, t_D) = -\frac{1}{2}\mathrm{Ei}\left(-\frac{r_D^2}{4t_D}\right)$$

Solution Step 7: Back to oilfield units

Substitute the definitions of p_D, r_D and t_D. The argument becomes

$$\frac{r_D^2}{4t_D} = \frac{\phi\mu c_t r^2}{4\,(0.0002637)\,kt} = \frac{948\,\phi\mu c_t r^2}{kt}$$

and the pressure solution is

$$p(r,t) = p_i + 70.6\frac{qB\mu}{kh}\,\mathrm{Ei}\left(-\frac{948\,\phi\mu c_t r^2}{kt}\right)$$

Ei(−x) is negative for x > 0, so p < p_i during production, as expected. The factor 70.6 is 141.2 divided by 2.

Solution Step 8: Logarithmic approximation

For small arguments the exponential integral has a series expansion:

$$E_1(x) = -\gamma - \ln x + x - \frac{x^2}{4} + \cdots, \qquad \gamma = 0.5772$$

When x = r_D²/(4t_D) < 0.01, the terms beyond −γ − ln x are negligible, so E₁(x) ≈ −ln(1.781 x). This holds when t_D/r_D² > 25. Then

$$p_D = \frac{1}{2}\left[\ln\frac{4t_D}{r_D^2} - 0.5772\right] = \frac{1}{2}\left[\ln\frac{t_D}{r_D^2} + 0.80907\right]$$

At the wellbore (r = r_w, r_D = 1), and converting to base-10 logs and field units, this gives the familiar semilog result:

$$p_{wf} = p_i - 162.6\,\frac{qB\mu}{kh}\left[\,\log t + \,\log\frac{k}{\phi\mu c_t r_w^2} - 3.23\right]$$

This is the basis of the **semilog (Horner/MDH) analysis**: a plot of p_wf against log t is a straight line whose slope gives permeability k.

Solution Step 9: Radius of investigation

The pressure disturbance is significant only where the Ei argument is of order one. Setting 948 φ μ c_t r² / (k t) = 1 gives

$$r_{inv} = \sqrt{\frac{k\,t}{948\,\phi\mu c_t}}$$

(t in hours, r in ft). It grows as the square root of time, which is the same √(η t) scaling noted for the linear case.

Solution summary

Step	What we did	Result
1	State PDE, initial and boundary conditions	Constant-rate line source, infinite reservoir
2	Make variables dimensionless	p_D, r_D, t_D
3	Boltzmann variable s = r_D²/(4t_D)	PDE becomes an ODE in s
4	Solve the ODE	f′ = C₁ e^(−s)/s
5	Inner boundary condition	C₁ = −1/2
6	Outer boundary and initial condition	p_D = −½ Ei(−s)
7	Back to field units	p = p_i + 70.6 (qBμ/kh) Ei(−948 φμc_t r²/kt)
8	Small-argument expansion	Semilog line, slope gives k
9	Where Ei argument is 1	Radius of investigation

Limitations: the line-source solution ignores wellbore storage and skin (important at early time) and boundaries (important at late time). Those effects are added on top of this solution in practical well-test analysis.

Notes: Common Questions and Nuances

N1. Why c_f for rock and c for fluid?

It is a notation convention. The **f stands for formation** (the rock), not fluid. Fluid compressibility carries a phase subscript in general: c_o (oil), c_w (water), c_g (gas). In the derivation there is a single fluid, so its compressibility is written simply as c. For multiphase flow the total compressibility is

$$c_t = c_f + S_o c_o + S_w c_w + S_g c_g$$

For one liquid phase this reduces to c_t = c + c_f, which is what was used.

N2. Why is porosity multiplied in the accumulation term?

Fluid is stored only in the pore space. The slice has bulk volume ΔV, but only the fraction φ of it is open pore space:

$$\text{pore volume} = \phi\,\Delta V, \qquad \text{mass stored} = \rho\;(\,\phi\,\Delta V\,)\;=\Delta V\,(\,\phi\rho\,)$$

Leaving φ out would treat the whole slice as filled with fluid, as if there were no rock.

Two related points:

- **φ is not in the mass in/out terms.** Darcy velocity v is flow rate per unit *total* (bulk) cross-sectional area, so porosity is already built into it.
- **φ sits inside the time derivative, ∂(φρ)/∂t.** Both ρ and φ change with pressure. Fluid expansion changes ρ (compressibility c). Pore volume change alters φ (compressibility c_f). The product rule on φρ is what produces c + c_f = c_t. Treating φ as constant would lose the rock compressibility.

N3. Is the derivation Cartesian?

- **The linear derivation is Cartesian.** Flow is along x, the slice has constant cross-section w·h, and the control volume is a rectangular block.

- **The radial derivation is cylindrical (r, θ, z).** The control volume is an annular shell and the flow area 2πrh grows with r. Nothing varies with θ or z, so only r matters.

Both are special cases of the general equation $\nabla^2 p = (1/\eta)\, \partial p/\partial t$, where the Laplacian depends on the coordinate system:

System	Laplacian $\nabla^2 p$
1D Cartesian	$\partial^2 p/\partial x^2$
2D Cartesian	$\partial^2 p/\partial x^2 + \partial^2 p/\partial y^2$
3D Cartesian	$\partial^2 p/\partial x^2 + \partial^2 p/\partial y^2 + \partial^2 p/\partial z^2$
1D radial	$(1/r)\, \partial/\partial r\, (r\, \partial p/\partial r)$
3D cylindrical	$(1/r)\, \partial/\partial r\, (r\, \partial p/\partial r) + (1/r^2)\, \partial^2 p/\partial \theta^2 + \partial^2 p/\partial z^2$

Only the geometry changes. The physics (mass balance, Darcy's law, compressibility) is identical.

N4. Where does R3 come from? (Differential vs integrated Darcy's law)

Darcy's law is the same in both geometries. The familiar form $q = kA\Delta p/(\mu L)$ is the **integrated** form, valid only for constant area and steady flow. The derivation uses the **differential (point) form**, which holds in any geometry and at any time.

Differential form (flow in $-r$ direction):

$$v = \frac{k}{\mu}\frac{\partial p}{\partial r} \qquad (R2)$$

Linear integrated form. Multiply by area, $q = vA = (kA/\mu)\, dp/dx$. With constant A and steady q, integrate over length L:

$$q = \frac{kA\,\Delta p}{\mu L}$$

Radial integrated form. Here $A = 2\pi rh$, so $q = (2\pi rhk/\mu)\, dp/dr$. Steady q is constant but r is not, so separate variables:

$$q\int_{r_w}^{r_e}\frac{dr}{r} = \frac{2\pi kh}{\mu}\int_{p_w}^{p_e} dp \quad \Rightarrow \quad q = \frac{2\pi kh\,(p_e - p_w)}{\mu\,\ln(r_e/r_w)}$$

A logarithm replaces the length L, but the differential form behind both is identical.

R3 itself is not Darcy's law alone. It comes from substituting (R2) into the radial continuity equation (R1). The geometry enters through R1 (the factor r from the area), not through Darcy:

$$r\rho v = r\rho\frac{k}{\mu}\frac{\partial p}{\partial r} \quad \Rightarrow \quad \frac{k}{\mu}\frac{1}{r}\frac{\partial}{\partial r}\left(r\rho\frac{\partial p}{\partial r}\right) = \frac{\partial(\phi\rho)}{\partial t}$$

The diffusivity equation describes **unsteady** flow, where q varies with position and time, so the steady-state integrated forms cannot be used. The integrated radial formula is the steady (or pseudo-steady) well inflow equation, which the transient solution approaches at late time in a bounded reservoir.

N5. The compact form in the radial linearization step

The claim is that the two expressions below are the same:

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r} \quad \text{and} \quad \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right)$$

Expand the compact form. Inside is a product of $a = r$ and $b = \partial p/\partial r$. The product rule, with $\partial r/\partial r = 1$, gives

$$\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right) = r\frac{\partial^2 p}{\partial r^2} + \frac{\partial p}{\partial r}$$

Multiply by 1/r. In the first term r cancels with 1/r:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right) = \frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r}$$

Numerical check with $p = r^3$. Expanded form: $p'' + p'/r = 6r + 3r = 9r$. Compact form: $r\,p' = 3r^3$, d/dr gives $9r^2$, times 1/r gives $9r$. Both agree.

Why use the compact form:

1. $r\,\partial p/\partial r$ is proportional to flow rate ($q = (2\pi kh/\mu)\, r\,\partial p/\partial r$), so the equation says "the change of flow rate with r equals the storage change".

2. The line-source boundary condition is a condition on that same quantity r ∂p/∂r.
3. It is the r-part of the cylindrical Laplacian.

The extra term (1/r) ∂p/∂r is the geometry correction from the growing flow area. Far from the well, 1/r is small and the radial equation looks more and more like the linear one.

N6. Why r is inside the derivative (radial) but x is not (linear)

What sits inside the derivative is the **flow area**, not r or x as such.

- **Linear:** the area w·h is the same at both faces, so it factors out of the difference and divides away. Nothing x-dependent remains.
- **Radial:** the area is 2π(r+Δr)h at the outer face and 2πrh at the inner face. Only 2π and h are constants. The factor r is different at each face, so it must stay inside the difference. After dividing by Δr the difference becomes a derivative of the whole product rρv.

Leaving r outside would drop a term:

$$\frac{\partial\,(\,r\rho v\,)}{\partial r} = \rho v + r\frac{\partial\,(\,\rho v\,)}{\partial r} \;\neq\; r\frac{\partial\,(\,\rho v\,)}{\partial r}$$

The extra ρv is the physical effect of the area growing with r. Check: for steady flow with constant q, rρv is constant, so ∂(rρv)/∂r = 0, giving ρv = C/r (velocity rises toward the well). Then ∂(ρv)/∂r is not zero, so keeping r outside would be wrong.

General rule for a flow area A(s) along the path s:

$$\frac{1}{A}\frac{\partial\,(\,A\rho v\,)}{\partial s} = \frac{\partial\,(\,\phi\rho\,)}{\partial t}$$

Geometry	Area A(s)	Constant?	Result
Linear	w·h	yes, cancels	∂(ρv)/∂x
Radial	2πrh	no	(1/r) ∂(rρv)/∂r
Spherical	4πr²	no	(1/r²) ∂(r²ρv)/∂r

A tapered linear channel with area A(x) would keep A(x) inside the bracket for exactly the same reason.

2D and 3D Cartesian Diffusivity Equation

The idea is the same as the 1D case, but fluid now crosses **three pairs of faces**. Here velocities are taken positive in the +x, +y, +z directions, so mass enters at the lower face of each pair and leaves at the upper one. The final result does not depend on this choice.

3D Step 1: Control volume and faces

A block with sides Δx, Δy, Δz has volume ΔV = Δx Δy Δz. Faces normal to x have area Δy Δz, faces normal to y have area Δx Δz, faces normal to z have area Δx Δy.

3D Step 2: Mass balance over Δt

Mass crossing each face is ρ A v Δt. Writing in minus out for each direction, and the accumulation Δx Δy Δz [(φρ)$_{t+\Delta t}$ − (φρ)$_t$]:

$$\Delta y\Delta z\Delta t(\,[\,\rho v_x\,]_{\,x} - [\,\rho v_x\,]_{\,x+\Delta x}) + \Delta x\Delta z\Delta t(\,\left[\rho v_y\right]_{\,y} - \left[\rho v_y\right]_{\,y+\Delta y}) + \Delta x\Delta y\Delta t(\,[\,\rho v_z\,]_{\,z} - [\,\rho v_z\,]_{\,z+\Delta z}) = \Delta x\Delta y\Delta z[\,(\,\phi\rho\,)_{\,t+\Delta t} - (\,\phi$$

3D Step 3: Divide and take limits

Divide by Δx Δy Δz Δt. In each term on the left, two lengths cancel and one stays in the denominator. Taking all increments to zero:

$$-\frac{\partial\,(\,\rho v_x\,)}{\partial x} - \frac{\partial\left(\,\rho v_y\right)}{\partial y} - \frac{\partial\,(\,\rho v_z\,)}{\partial z} = \frac{\partial\,(\,\phi\rho\,)}{\partial t} \qquad (\,3D\text{-}1\,)$$

In vector form this is −∇·(ρ**v**) = ∂(φρ)/∂t.

3D Step 4: Darcy's law in each direction

$$v_x = -\frac{k}{\mu}\frac{\partial p}{\partial x}, \qquad v_y = -\frac{k}{\mu}\frac{\partial p}{\partial y}, \qquad v_z = -\frac{k}{\mu}\frac{\partial p}{\partial z}$$

Substituting into (3D-1), the two minus signs cancel and k, μ come out of the derivatives:

$$\frac{k}{\mu}\left[\frac{\partial}{\partial x}\left(\rho\frac{\partial p}{\partial x}\right)+\frac{\partial}{\partial y}\left(\rho\frac{\partial p}{\partial y}\right)+\frac{\partial}{\partial z}\left(\rho\frac{\partial p}{\partial z}\right)\right]=\frac{\partial\,(\,\phi\rho\,)}{\partial t}$$

3D Step 5: Expand and linearize

Each term expands as in 1D (product rule, then ∂ρ/∂x = cρ ∂p/∂x):

$$\frac{\partial}{\partial x}\left(\rho\frac{\partial p}{\partial x}\right)=\rho\frac{\partial^2 p}{\partial x^2}+c\rho\left(\frac{\partial p}{\partial x}\right)^2$$

and similarly for y and z. The accumulation term is unchanged, ∂(φρ)/∂t = φρ c_t ∂p/∂t. Cancelling ρ and multiplying by μ/k:

$$\frac{\partial^2 p}{\partial x^2}+\frac{\partial^2 p}{\partial y^2}+\frac{\partial^2 p}{\partial z^2}+c\left[\left(\frac{\partial p}{\partial x}\right)^2+\left(\frac{\partial p}{\partial y}\right)^2+\left(\frac{\partial p}{\partial z}\right)^2\right]=\frac{\phi\mu c_t}{k}\frac{\partial p}{\partial t}$$

Dropping the small c|∇p|² term gives the 3D diffusivity equation:

$$\frac{\partial^2 p}{\partial x^2}+\frac{\partial^2 p}{\partial y^2}+\frac{\partial^2 p}{\partial z^2}=\frac{1}{\eta}\frac{\partial p}{\partial t},\qquad \eta=\frac{k}{\phi\mu c_t}$$

2D case

For a thin reservoir of thickness h with flow only in the x-y plane, the block is Δx × Δy × h. The thickness h appears in every term and cancels, just as w·h did in 1D. The result is

$$\frac{\partial^2 p}{\partial x^2}+\frac{\partial^2 p}{\partial y^2}=\frac{1}{\eta}\frac{\partial p}{\partial t}$$

Two practical notes

Anisotropy. If permeability differs by direction, Darcy's law uses the matching permeability in each direction:

$$k_x\frac{\partial^2 p}{\partial x^2}+k_y\frac{\partial^2 p}{\partial y^2}+k_z\frac{\partial^2 p}{\partial z^2}=\phi\mu c_t\frac{\partial p}{\partial t}$$

Vertical permeability k_z is usually the smallest.

Gravity. In the vertical direction Darcy's law has an extra term, v_z = −(k/μ)(∂p/∂z + ρg) with z measured upward. It was neglected above. For a slightly compressible liquid, ρ varies very little, so the gravity term is nearly constant in space and drops out of the derivative at this level of approximation. Pressure can then be read as pressure referred to a datum. For accurate vertical-flow problems (gas caps, coning), flow potential is used instead of pressure.